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Hybrid deep learning based intelligent security architecture for web threat mitigation in mobile communication networks

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Abstract

In the current artificial intelligence era, mobile network channels play a crucial role in the development of networks with 4G, 5G, and next generation 6G. This research focuses on intellectual security architecture (ISA) to mitigate web-oriented attacks that use hybrid deep learning models to stall the operation of mobile networks. With the rapid evolution of mobile communication systems, especially in the context of edge-based content sharing, network security has become a crucial aspect of maintaining robust and reliable service. The integration of artificial intelligence (AI) into security frameworks has opened new frontiers in identifying and mitigating cyber threats such as web-based attacks. This research suggests a deep learning system that uses lightweight bidirectional long short-term memory (Bi-LSTM) networks and transformer-based multi-head attention (MHA) to smartly detect web attacks. Apart from dataset accuracy, the proposed ISA is assessed for edge feasibility through testing. The ISA architecture requires 30,087 parameters and 0.42 MB of storage with a medium inference latency under 90 ms, thereby supporting on device deployment on limited storage 6G edge node. The proposed model is evaluated on a labeled web traffic dataset and trained using an additional zero shot, cross-dataset evaluation on independent edge IIoT dataset. The proposed model preserves effective attack/normal separability AUC of 0.93% without any domain specific fine tuning. Additionally, the proposed ISA model is very good at predicting different web attacks such as brute force, SQL injection, and background shell attacks, achieving an accuracy of 97%, while other models like multioutput Cat-Boost and Char-LSTM achieved accuracies of 88% and 85%, respectively. The proposed hybrid model performs well in predicting web attacks which indicates the learned representation generalized beyond its training distribution and ISA is suitable for lightweight and federated adaption on edge devices.

Keywords Intellectual security architecture (ISA) · 6G networks · Bi-LSTM · Transformer-based MHA · Web attack mitigation · Deep hybrid intelligence

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Abbreviations

ISA	Intellectual security architecture
AI	Artificial intelligence
Bi-LSTM	Bidirectional long short-term memory
MHA	Multi-head attention
IIoT	Industrial internet of things
ML	Machine learning
DL	Deep learning
CNN	Convolutional neural networks
HTTP	Hypertext transfer protocol (HTTP)

1 Introduction

As 6G networks process toward edge deployment, huge machine type communication, and network slicing, the web attacks surface for web-oriented threats increases rapidly. Edge nodes process latency with limited security instrumentation where data are increasingly served at the network edge rather than centralized data centers. Safeguarding from web-based attacks such as brute force, SQL injection and shell script exploitation with this setting requires a good architecture that is accurate, lightweight enough for edge hardware and validate the ability to generalize across traffic domains.

The increase in use of mobile devices have led to the rise of dependency on mobile networks and in as a result escalated web-based attack. With the emergence of next generation 6G networks and the growth of edge-based content sharing, security challenges are amplified. According to the report of Edge-scan, mobile network-based vulnerabilities and web attacks have surged drastically since the year 2023. Among the most well-known web attacks are brute force, shell script, and SQL injection. As mobile traffic continues to dominate the internet usage globally [1], protecting these networks, especially in the context of 6G edge deployments, has become very important. The mobile environment not only have increased the attack surface in traditional web attacks [2] but also have opened new security challenges related to distributed edge computing [3]. Deep learning (DL) techniques are being applied across various cybersecurity areas like infrastructure threats and attacks, software attacks, and privacy preservation [4]. However, these approaches often struggle with the complex, multi-dimensional nature of modern web attacks targeting mobile infrastructure and 6G edge resources. Researchers have explored the application of deep learning methods, to detect anomalies in network traffic and identify potential security breaches [5, 6].

To solve these problems and establish a robust intellectual security architecture (ISA), this paper introduces a new mixed design that combines bidirectional long short-term memory (Bi-LSTM) with a transformer-based multi-head attention (MHA) mechanism. Bi-LSTM is well-suited for analyzing network request patterns as it can capture long-term dependencies in sequential data [7]. The main concept of the lightweight transformer is to enable a model to focus on information from different areas of representation at the same time. In the context of ISA and network security, this capability can be valuable for analyzing various aspects of network traffic and behavior concurrently. Mobile communication environments constantly analyze the types of attacks that process inside the web application that are used to monitor the channel and user activity inside the network. Traditionally, all domains integrate artificial intelligence models to increase the efficiency of the processes and achieved high accuracy. Some traditional deep learning models like LSTM, convolutional neural networks, and artificial neural networks are utilized to analyze the web application that is used in the mobile communication organization, but training this model takes more time, some predictions are lower, and additional resources are needed to get perfect output. To overcome the above issues, this research used the hybrid model by integrating two models: Bi-LSTM and a lightweight transformer model named the multi-head attention mechanism.

The hybrid model has more advantages than the standalone deep learning (DL) model. Training deep learning models, which needs many resources like GPUs for additional support in training and predicting the outcome with

high accuracy. In the proposed research, deep learning model namely Bi-LSTM is used. This model is trained using forward and backward passes to analyze the long-term dependency pattern inside the mobile network. After capturing the dependencies, it is forwarded to lightweight transformer model that analyze the whole information inside the mobile network to capture different types of attacks processed inside the mobile communication networks. Malicious activities including brute force, shell scripting, and injection attack are identified efficiently with high accuracy. The lightweight prediction model will not require much resources for training or any additional GPU for testing with particular dataset. Many existing research uses machine learning models to predict web attacks, but those are efficient in smaller datasets. In this study, the proposed hybrid model combines two efficient models that are trained on large datasets, allowing for quick predictions with high accuracy.

Developed in the year 2016, the available dataset to detect web attacks is UNSW-NB 15, suited to real-time data processing with different types of attacks. However, it is an unbalanced dataset and some attacks cannot be detected. Similarly, CICDDoS-2019 dataset primarily focuses on DDoS attacks and is functions effectively in IoT-type device environments, making it well-suited for DDoS attacks but not for other types of attacks. Another dataset is IoT-23, which contains data related to malware details in the IoT environment. As the raw data is structured in JSON, the preprocessing required for feature extraction is complex and challenging, even though the dataset itself is highly suited for malware detection tasks. The next dataset is TON-IoT0, which contains the details of log data of wireless networks and NetFlow data from some industrial IoT device, but this dataset is hard to preprocess. Based on the above limitations of recent and old datasets, this research uses the CICIDS2017 dataset [8]. While it is relatively old, but it contains various types of attacks in realistic traffic data. Its well-structured contents make it cover broad attack coverage and ensure better modeling for interaction patterns between normal and web-oriented functions.

To develop a solution to address these issues, a robust model is required. In existing literature, it is a combined contribution of Bi-LSTM and transformer-based multi-head attention hybrid model for web attack detection in mobile edge deployments. The contribution of this research is (i) a controlled ablation quantifying the accuracy gain attributable to the hybrid combination over each component alone, (ii) a system level model establishing edge deployment feasibility, and (iii) a cross-dataset evaluation on an independent edge IoT dataset to perform stress testing generalization which is limited in existing hybrid IDS studies.

1.1 Main contribution of the research

The goal of this research work is

- This research introduces a novel intellectual security architecture (ISA) that combines a lightweight Bi-LSTM temporal encoder with a transformer-based MHA model, specifically designed for detecting web attacks.
- To conduct a controlled ablation study separating the accuracy contribution of each component like Bi-LSTM only, MHA only, and hybrid model quantifying the benefit of combing them in attack detection.
- To provide a system level characterization such as parameter count, disk footprint, inference latency, and throughput for edge deployment feasibility.
- To perform exhaustive evaluation using a diverse dataset on trained model zero shot on an independent edge IIoT dataset reporting its strength as preserved AUC and limitation reduced accuracy on domain shift.
- An analysis of model performance with class weighting and loss configuration to handle severe imbalance cross seven traffic categories to support reproducibility.

1.2 Organization of the paper

This research is organized into several sections: Sect. 2 reviews existing studies on web attack detection and intelligent security architectures (ISA) in mobile and 6G edge networks; Sect. 3 explains the methods and materials,

including the details of the proposed intellectual security architecture (ISA) and the hybrid Bi-LSTM and MHA model algorithm architecture; Sect. 4 presents the results and experiments showing how the proposed work is better than traditional models; Sect. 5 delivers with discussion and comparison of results; and Sect. 6 wraps up with conclusions and future plans.

2 Literature review

In recent years, the detection of web attacks in mobile networks is attracting the attention of the researchers. To enhance security in mobile networks, and specifically in the context of edge-based content sharing in next generation 6G networks, numerous ML and DL approaches have been proposed in the literature.

Tian et al. [9] proposed a hybrid model which combines long short-term memory (LSTM) and convolutional neural networks (CNNs) for web attack detection. To identify the malicious patterns, they processed HTTP request as character sequences and then extracted both the temporal and spatial features. Their approach demonstrated 98.2% accuracy with a false positive rate of 0.76% that was tested against five types of web attacks. Although while dealing with unknown attack types like zero-day attack, their accuracy reduced to 12% which shows the limitation of the model while dealing with the dynamic mobile environment.

Zhang et al. [10] introduced a network intrusion detection system (NIDS) combining Bi-LSTM and multi-head attention mechanism. An embedding layer was used to convert input features into dense vector and then the multi-head attention assigned different weights to various parts of the input sequence in a way enhancing the focus of the model on critical features. They evaluated their model on datasets namely KDDCUP99, NSLKDD, and CICIDS2017 that recorded accuracies of 98.29%, 95.19%, and 99.08%, respectively. Due to the inherent properties of mobile environments, particularly resource constraint and real-time processing capabilities (critical in edge deployments), the proposed model's applicability to mobile networks remains unexplored.

Xiaolong et al. [11] proposed a character-level Bi-LSTM with multi-head model aiming cross-site scripting (XSS) attacks within cloud computing platforms. The model could effectively capture the bleary malicious scripts by using the character-level embedding to overcome the word-level representation limitations. Results indicated significant improvement in detection with F1 score of 98.71%. The model did not address the performance in mobile network scenarios. Mobile have limited computational resource and latency related issues that might hamper the proposed character-level processing which is already burdened with additional computational overhead.

Jouhari et al. [12] presents a combined Bi-LSTM and CNN, IDS specifically for IoT devices. In their hybrid architecture, CNN extracts the spatial features of the input data and the Bi-LSTM captures the temporal dependencies, easing the task of analyzing sequential patterns efficiently. The model achieved accuracies of 97.28% for binary classification and 96.91% for multiclass scenarios. Although the model is suitable for devices with limited computational capabilities, they do not extend their work to mobile network conditions or 6G edge environments. It would be beneficial to access the model's performance under real-time constraints and its adaptability to the dynamic nature of mobile network traffic.

Rachid et al. [13] addresses the issue of NIDS in software-defined networking (SDN) environments, using a hybrid DL model. The combined CNN Bi-LSTM networks include balancing datasets with random oversampling and feature selection using a random forest classifier. They achieved around 97.77% accuracy on the InSDN dataset for multiclass classification. The approach indeed, addresses the challenges, the data redundancy, and unbalanced datasets that are common to NIDS and mobile networks. They focused on specific dataset affecting generalized adaptability to diverse mobile environment and edge architectures.

Gao [14] combined CNN and Bi-LSTM in cloud computing environment for NIDS. They used data extraction algorithm for preprocessing the data and transformed it into an image dataset. The experimental results showed that the proposed method achieved an average accuracy of 95.50%, while the false positive rate and the false negative is reduced to 4.24% and 6.66%, respectively. The paper does not specifically address the limitations of the proposed method in the context of mobile networks or 6G.

Zahoora et al. [15] proposes a model for ransomware detection using DL for unsupervised feature extraction that focuses on detecting zero-day attacks. The framework may be too complex for real-time processing on mobile devices/edge nodes. They do not address the challenges of applying the CSPE-R to the diverse operating systems and architectures of mobile devices.

Redhu et al. [16] presents how DL approaches can improve detection accuracy and address challenges posed by malware in cyberspace. They discuss how traditional security techniques fail to detect evolving malware. Among the various DL models reviewed such as CNN, LSTM-based and autoencoder-based, the best detection accuracies ranged from 96 to 99.5%. They acknowledge that DL models are susceptible to adversarial attacks and model interpretability is still a challenge. Due to resource constraint factor in mobile devices (edge nodes), computationally intensiveness poses a challenge for real-time deployment. Latency issues, energy consumption, and data privacy are few other factors that need to be looked upon to be applicable to mobile network environments (like 6G).

Haoyu et al. [17] presents a proactive cybersecurity mechanism to detect anomalies in near real time. Their model is based on DL framework designed to detect early-stage large scale network attacks. Net Sentry shows an accuracy of 98.79%, precision of 98.80%, recall of 98.75%, and F1 score of 98.77% in discovering attacks, demonstrating its robustness and high efficacy in differentiating between normal and malicious activities in network. The authors accepts that challenges like increased noise, user mobility, variable latency, and bandwidth limitations of mobile and wireless network (6G) may affect detection accuracy. They suggest more adaption to their model to maintain accuracy in such heterogeneous conditions of mobile environment.

Recently, attention methods have become effective instruments for security applications [19]. [20] revolutionized the sequence modeling by introducing the transformer architecture with multi-head attention. It achieves state-of-the-art BLEU scores of 28.4 on English-to-German and 41.0 on English-to-French translation tasks. The transformer model lacks quadratic time complexity with respect to sequence length and struggles in handling very long sequences efficiently. In context to web security, attention mechanism was used along with DL models for detection of SQL injection [21] and in [13] for network anomaly detection. [22] in their paper highlighted the potential of attention-based mechanism for designating security threats in encrypted mobile traffic. The most effective models reached about 85% accuracy when distinguishing between multiple mobile apps. Crucial limitations are sensitivity to app version changes, reduced effectiveness with very similar app traffic and reliance on carefully chosen input formats. Because, very few studies have used attention mechanism for mobile security, the development of the proposed ISA is well justified.

According to recent literature, hybrid architectures that combine traditional approaches produce better outcomes for challenging security problems. The effectiveness of federated learning with an auto encoder-based convolutional network for intrusion detection in consumer electronics is discussed in [23]. Dinushan Sivaneswaran et al. [24] explored a PRISMA methodology that utilized LLM architecture to identify the phishing attacks processed by the attackers. The proposed model showed high classification accuracy for analysis of textual and visual content for phishing detection, outperforming many similar existing models.

Even with these developments, there is still a significant gap in how the unique problems of web attack detection in mobile environments, specifically in 6G edge-based content sharing. Specialized methods that may preserve high detection accuracy while staying computationally economical are required due to the distinct features of mobile networks, such as resource limits, varied traffic patterns, and bandwidth variability [25, 26]. According to above literature from [9–26], hybrid models like CNN/Bi-LSTM/attention architecture consistently perform good with high accuracy on web-based attack detection on traffic and 6G-based IIoT dataset ranging from 95 to 99%. The results confirm that combining temporal and attention-based feature extraction is very effective for web attack detection. The common research gap arises however mentioned as (1) Almost existing work not mostly reviewed cross-data generalization model and evaluated on same distribution, (2) System level evaluation like latency, memory, and energy are very less report rather than frequent claims of edge and lightweight suitability,

and (3) mobile, edge, 6G traffic conditions are processed as future work not reviewed rather than test directly in the research. The proposed intellectual security architecture (ISA) addresses this gap.

3 Research gap

The above-mentioned literature classifies the following methodological gaps, which are particularly pronounced in next generation 6G and edge-based content sharing environments:

(1) Many traditional models of web attack detection are based on heavy and deeper architecture models, which are not suitable for edge devices and resource-constrained computing environments.
(2) Current deep learning models are slow in detecting real-time web attacks and provide overfitting results with high classification accuracy on imbalanced multiclass attack datasets, making them unsuitable for the high-speed, dynamic nature of 6G traffic.
(3) Most models are black boxes that do not consider important data such as HTTP requests and URL mapping; this results in poor interpretability of the predicted outcomes, which is vital for forensic analysis within an Intellectual Security Architecture (ISA).

This gap is bridged by introducing a novel hybrid architecture, the Intellectual Security Architecture (ISA), designed especially for 6G mobile environment and edge deployment. Our method combines the strengths of Bi-LSTM, which tracks changes over time, and Transformer-Based Multi-Head Attention, which focuses on important features from different perspectives, to ensure high detection accuracy and computational efficiency for mobile networks.

4 Methods and experiments

The proposed research is to predict the different types of web attacks in next generation 6G mobile communication networks, which may stall the efficiency of mobile networks with coverage issues and low bandwidth in metropolitan areas, especially affecting edge-based content sharing. To effectively study the various types of attacks in mobile communication, the suggested intellectual security architecture (ISA) model was tested using a real-time dataset and methods to prepare the data, identify important features, and ultimately detect the different types of web attacks in the network [27].

4.1 Material

4.1.1 Dataset description

The proposed work uses the CICIDS2017 dataset for testing. The CICIDS2017 dataset is special because it includes labeled network flows that show different kinds of web attacks, such as brute force, shell scripts, and SQL injection, unlike some newer datasets. This dataset replicates the real-time realistic testing of types of web attacks in mobile communication networks, providing traffic patterns relevant to edge-based content sharing. These attacks are processed with hypertext transfer protocol (HTTP)-based attacks for decreasing the efficiency of mobile networks. To protect the efficiency of mobile networks, these attacks are predicted early to allow the intellectual security architecture (ISA) to access the resource efficiently. Table 1 illustrates the dataset parameters used for testing [28, 29].

Table 1 Dataset parameters

Features	Testing data
Traffic data	Source and destination IP
Headers	HTTP, URL
Payload	Content, query, token
Network info	Port, packet size, duration
Training and validation set	70:30
Label	Bots, DDoS, DOS, Port scanning, Web attacks-Brute force, SQL injection

4.1.1.1 Data preprocessing The data processing is carried out by executing different processes with the available dataset. This is a crucial step for preparing network flow data, particularly for the high-speed and varied traffic of 6G edge networks. The initial process is to align and clean the dataset, the input data are then processed with setting labels and features. Source and destination IP addresses associate the labels, while the vector form fixes the ranges. Next, the data are processed with a number of samples with different numbers of features and different numbers of classes. Equation (1) defines the initial input data processed for training [32].

$$ID = (x_i, y_i) \tag{1}$$

where ID as input data, x and y are labeled data for each entering internet protocol (IP) address, i as ranges limit of devices connected in the network.

Next step is scaling, that align the data by normalizing its values based on the median and data range represented in vector form as i, j. The scaling is to find the average number of data processed into the proposed model for training. Equation (2) expresses this scaling.

$$scale = x_{ij} - median/range\left(x_{ij}\right) \tag{2}$$

After figuring out the scale value to change the median, different network flows are found for the next layer by using label encoding to sort the labeled data and get rid of duplicates. Removing duplicates from large, time-stamped HTTP text requests is essential for reducing dataset noise and ensuring effective model training. After removing duplicates from the dataset, the data are then sent to feature extraction in the proposed model with various layers to improve training for predicting different web attacks. Equation (3) expresses the label encoding equation [33].

$$LE = LE(y_{ij}) \tag{3}$$

where LE is label encoder, yij are vector data for different network flows in the mobile communication network.

4.1.2 Cross dataset generalization protocol

To evaluate the applicability beyond conventional data traffic and address the mismatch between the CICIDS2017 and true 6G edge condition, the trained ISA architecture and its single-branch variant are further evaluated in a zero shot strict settings on the edge IIoT dataset without retraining and threshold recalibration. The dataset specifically captures IoT-based edge device traffic more suitable for resource-constrained edge nodes. This evaluation protocol enables lower bound on deployable cross domain performance prior to the federated fine tuning of ISA architecture hybrid model.

4.2 Proposed methodology

The proposed model uses lightweight deep learning integrated with a transformer-based multi-head attention (MHA) model to predict web attacks in the 6G mobile communication network. The model uses bidirectional long short-term memory (Bi-LSTM) to understand the order of events by looking at the data from both the past and the future in the mobile communication network traffic. After the Bi-LSTM identifies the time-related patterns, this information is sent to the next model, MHA, which looks at the attention weights for each time period of attacks, helping to clearly recognize attack patterns. The lightweight nature of the proposed model is integral to the intellectual security architecture (ISA), allowing for easy deployment in edge devices, where it monitors 6G mobile communication infrastructure and provides a dashboard for attack prediction. The below section defines the system model and architecture of the proposed work. Figure 1 defines the schematic diagram of the proposed methodology.

4.2.1 Bidirectional long short-term memory

The initial layer of the intellectual security architecture (ISA) starts feature extraction after preprocessing the dataset's initial data for proper training. The proposed model is lightweight and uses 64 hidden layers and optimized regularization. Through this, the model learns complex temporal patterns in the network with irregular traffic inside, and some unauthorized access is noted as web attacks like brute force, shell scripts, and SQL injection. The initial process of the Bi-LSTM, which includes an input layer, accepts data from 6G edge communication network activities. The whole process processed by Bi-LSTM is expressed below.

a. Input Layer: In this layer, the total number of time steps calculated for different data like IP address, packet size, and duration are processed in a sequence. Equation (4) expresses the input sequence.[34]

$$I = \{x_1, x_2 \ldots .x_{TS}\} \tag{4}$$

where I is the input sequence, x is the individual sequence, and TS denotes the time steps calculated from the packet and duration for each user in the network.

b. Forward Pass: This layer is processed after the input layer is processed with a gate function to analyze the sequence of data. The layer will have a sigmoid activation function as the initial activation function with a bias

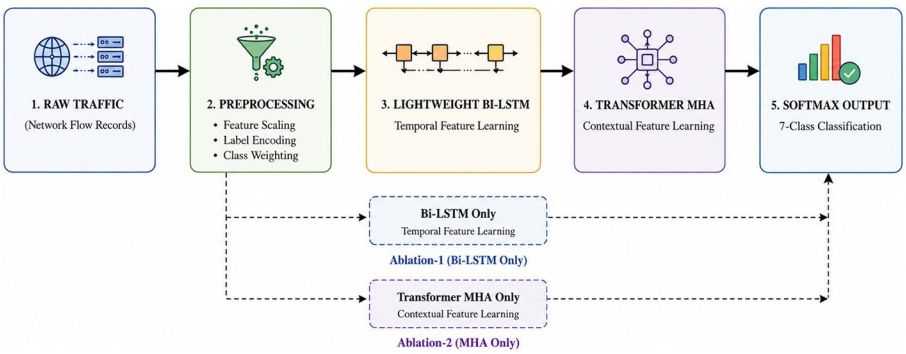


Fig. 1 Schematic flow of the proposed deep learning framework with multiple processing layers. The figure illustrates the complete architecture of the proposed web attack detection framework. The workflow begins with acquisition of network traffic records from the benchmarked dataset followed by data preprocessing, feature normalization, categorical encoding and feature selection. The processed feature is the provided to the Bidirectional long short-term memory (Bi-LSTM) network for extracting temporal dependencies from sequential traffic patterns. A multi-head attention (MHA) mechanism is subsequently employed to identify the most informative feature contributing to attack classification while supressing irrelevant information

vector, where weight matrices are processed to analyze the time of request inside the mobile communication network. Equation (5) expresses the forget gate f_p.

$$f_p = \sigma(W_f x_t + U_f + b_f) \tag{5}$$

where f_p represents the forward pass, σ is the initial sigmoid activation, W and U (implicitly included in W in the general form) are weight matrices, and b_f as bias.

c. LSTM Input Layer: This layer process the output from the forward pass forwarded from the gate then processing done with next level layer to capture that output into next input layer. Equation (6) expresses the input gate i_t.

$$i_t = \sigma(W_i x_t + U_i + b_i) \tag{6}$$

where i_t represents the next input of LSTM as input gate function.

d. Cell Update: This layer processes the memory of the LSTM input function generated which processes the data from the forward pass. This layer processes with the next activation function to note the sequences with the hyperbolic tangent activation tanh. Equation (7) expresses the candidate cell update CU_t. [35]

$$CU_t = \tanh(W_c x_t + U_c + b_c) \tag{7}$$

where CU is the candidate cell update, and c relates to the candidate sequence class.

e. Cell State: This layer processes the outcome from the candidate updates which moves toward the next memory in gate, after process an update the sequence is further extracted for further processing with element-wise multiplication Θ. Equation (8) defines the new cell state C_t.

$$CS_t = f_t \odot c_t + i_t \odot \overline{CS_t} \tag{8}$$

where C_t is the cell state, ⊙ is the element-wise multiplication processed inside the gates.

f. Backward Pass Output: This layer processes the output from the final memory update by analyzing all the sequence and from general HTTP request and extracting exact details from the URL. The formula is expressed in Eq. (9) for the output gate o_t. [36]

$$o_t = \sigma(W_o x_t + U_o + b_o) \tag{9}$$

where o_t represents the output from the gate.

g. Hidden States: After output passes from the backward pass to the next hidden state of Bi-LSTM, where it uses 64 hidden layers to process the output related to time. Equation (10) expresses the hidden state hs_t.

$$hs_t = o_t \odot \tanh(c_t) \tag{10}$$

where hs_t represents the hidden state which analyses the duration related to time t.

h. Bi-LSTM Concatenation: From the hidden state layer, the output is then processed to map the temporal dependencies like port utilization, and data access points are noted with the help of both forward h_t) and backward h_t) hidden state passes. Equation (11) expresses the final output concatenation for the Bi-LSTM layer.[37]

$$Concat = [\overrightarrow{h_t};\overleftarrow{h_t}] \tag{11}$$

where $Concat$ is concatenation related to both forward and backward pass gate sequence analysis function, producing the final feature representation which is then passed to the MHA layer.

4.2.2 Transformer-based multi-head attention mechanism

The transformer-based multi-head attention (MHA) takes in all the sequential information, like port scans and attack accessor IPs, from the last output of the Bi-LSTM mapping. This process of integrating the Bi-LSTM output with MHA is a core part of the intellectual security architecture (ISA). The MHA primarily concentrates on predicting attack-type sequences through contextual awareness mapping, utilizing attention weights.

One advantage compared to Bi-LSTM is that the transformer-based MHA captures long-range dependencies more easily compared to LSTM and other deep learning models, which is crucial for detecting subtle, multi-stage attacks in high-latency 6G networks. The MHA allows correlation between distant events in a long sequence of data inside the mobile communication network. The sequences include time exploits, such as brute force attacks, and patterns, such as shell scripts. All modes for transferring pattern data are noted inside the network by predicting web attacks and safeguarding the mobile communication network [38].

4.2.2.1 Query key value Query Key Value: The first layer in the MHA process is to map the sequence with their important function; they are query, key, and value. The query is to analyze the feature processed by the input taken from the final layer of Bi-LSTM; key values are fixed for each weight matrix where transformation of data is processed. Finally, the value vector is used to map with the attention weights further to predict the final output, which is used by the intellectual security architecture (ISA) for prediction. Equation (12) expresses the formula for focusing attention.

$$Q = HW^{Q}, K = HW^{K}, V = HW^{V} \tag{12}$$

where W is expressed as weighted projection matrices used for the transformation.

4.2.2.2 Dot product attention After using the attention function as the first step to change the dimensions of each head from the initial layer to capture long-range dependencies, create a foundation for dot product attention, shown in Eq. (13) [39].

$$Attention(Q, K, V) = Softmax\left(QK^{T}\Big/\sqrt{d_k}\right)V \tag{13}$$

where d_k are the dimensions of key vectors inside the attention layer with activation function.

4.2.2.3 MHA output The MHA output is used to identify the complex attack pattern with different types of relationships to concatenate each head attention from the query, key, and value. Each head is having its own projection to map to predict the final attack type, which is expressed in Eq. (14) [40].

$$MHA(Q, K, V) = Concat\left(head_1, \ldots head_h\right).W_o \tag{14}$$

where concatenation expressed as *Concat* for each head, and W_o denotes the final projection output.

4.2.3 System model

The final system model is to classify which types of web attacks are predicted with accuracy by using residual connections, normalization, pooling, loss calculation, and final prediction. All the mentioned functions are elaborated briefly below [41].

After the model uses temporal dependencies (from Bi-LSTM) and contextual awareness attention (from MHA), the data from the last output layer of MHA are sent through a residual connection to improve training stability and address the vanishing gradient problem. The residual connection helps solve the gradient issue and keeps track of long sequences of information for the final classification, which is crucial for maintaining performance on long 6G traffic flows [42]. The residual connection process enables overcoming the gradient problem and storing the long dependency sequence of information for the final classification. The equation for the residual connection is expressed in Eq. (15):

$$RC = H + MHA(H) \tag{15}$$

where RC refers as residual connection, H as output from the final layer.

After creating a residual connection to solve the vanishing gradient problem, the next step in the intellectual security architecture (ISA) is to use normalization in all layers to handle deeper models without overfitting. The normalization enhances the learning capability of the proposed model by increasing the speed of training with large or limited training samples to attain intended results, which is vital for efficient deployment on 6G edge nodes. [43] The normalization process considers all the token embeddings with fair distribution of multiple attentions in eq.(16):

$$\widehat{RC} = Norm(RC) \tag{16}$$

where RC denotes the inverse dimension of sequence, norm represents the normalization and average function defined in Eq. (17).

$$RC_{avg} = \frac{1}{T}\widehat{RC} \tag{17}$$

where T denotes the sequence length, and $\widehat{RC}$ represents the timestep output from normalization.

The next process after normalization is the pooling function, which reduces the length of each sequence, like link address, HTTP request, and duration, to a fixed length. This is important for creating compact feature vectors for the final classification layer within the intellectual security architecture (ISA) [44]. The average pooling function removes the dimension of time to make each sequence feature to enter the next dense layer as a fixed-size feature vector. The equation of average pooling is expressed in Eq. (18).

$$RC_1 = ReLU(W_1 RC_{avg} + b_1) \tag{18}$$

where w is the weight matrix, and ReLU represents the activation function to handle negative values and capture complex attack patterns.

The mobile communication network processes different classifications of web attacks using the next layer. The final activation sigmoid is used to predict the web attack based on the probability of the attack sample, where 1 is normal and − 1 is an attack. This step within the intellectual security architecture (ISA) provides the final decision on the nature of the network flow.[45]

$$\hat{y} = \sigma(W_2 RC_1 + b_2) \tag{19}$$

where σ is the output probability of the sigmoid activation, and W is the weight matrix of second layer.

Finally, binary cross-entropy loss is used to train the model in an effective way to handle model loss and prevent overfitting. The main advantage of this loss function is its ability to minimize classification errors related to web attacks or normal operations within the mobile communication network, which is essential for the reliability of the intellectual security architecture (ISA) in 6G edge nodes. [46] The final layer applies a SoftMax activation over seven output classes such as normal traffic, bots, brute force, DDoS, DoS, Port scanning, and web attacks process a variable distribution over attack types expressed in Eq. (20).

$$\hat{y} = \text{Soft max}\left(W_2 RC_{avg} + b_2\right) \tag{20}$$

Given the severe class imbalance across the seven categories, the model is trained with class weighted sparse categorical cross-entropy expressed in Eq. (21).

$$\text{Loss} = -\sum_{c=1}^{7} w_c y_c \log\left(\hat{y}_c\right) \tag{21}$$

where w_c is the weight applied to class c, computed inversely proportional to class frequency and capped at a maximum of 10 to prevent up weighting of rare classes from destabilizing training the weights used were: bots with weight as 10 (capped), brute force with weight as 10 (capped), DDoS with weight as 2.81, DoS with weight as 1.86, normal traffic with weight as 0.17 (high number of class), port scanning with weight as 3.97, and finally web attacks with weight as 10 (capped).

This process allows the hybrid multimodal to learn from its overfitting classification error and gradually improve performance during testing and validation. Training used the Adam optimizer, initial learning rate 0.001, reduced on plateau factor as 0.5 when validation loss stagnated up to 30 epochs mixed precision of float 16 training. The full configuration is reported to support reproducibility.

4.2.4 Architecture and algorithm

The proposed work architecture is to project the output for detecting web attacks that happen inside the mobile communication network. The initial layer is to take input from the dataset and then extract it as a network traffic feature. The next layer is the proposed hybrid multimodal (ISA) used to predict the web attack. The initial process involves capturing the temporal dependencies of the data by utilizing forward and backward passes with Bi-LSTM. MHA analyzes inner features for contextual attention mapping using a lightweight transformer-based multi-head attention. Once the model extracts the features, it uses residual connections to stabilize and improve the model's learning [47]. The next step involves normalizing the activation and then passing it to the pooling function, which generates a fixed length sequence for easier data validation. Lastly, the output layer classifies the output as either attack or normal and then calculates the prediction classification accuracy to validate the proposed work. Figure 2 defines the conceptual model of the proposed intellectual security architecture (ISA) model.

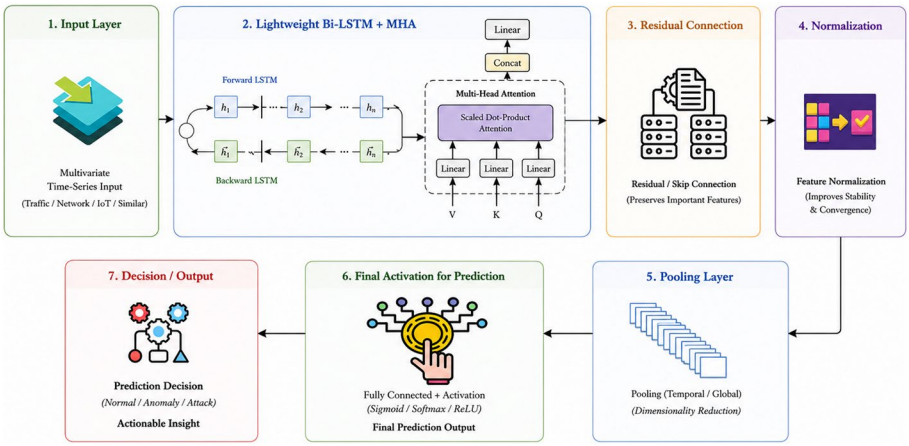


Fig. 2 Conceptual Visualization of the proposed intelligent attack detection framework. The figure illustrates the conceptual operation of the proposed Bi-LSTM with multi-head attention framework for intelligent web attack detection. The incoming network traffic is first passed into structured feature vectors through preprocessing and normalization. The Bi-LSTM model captures bidirectional temporal dependencies within the traffic sequence while the attention module assigns adaptive weights to discriminative features that contribute most significantly to attack classification. The refined feature representation is subsequently utilized by the classification layer to detect different categories of web attacks

INPUT: X as input of network flow traffic data from the dataset

OUTPUT: Predicted label as normal or Web Attack

Begin
Load the CICIDS dataset
Initilaize
Number of hidden layers in Bi-LSTM
Number of heads in Transformer based MHA
Weight matrices as Q K V
For each head **do**
$Weight\ bias\ for\ dense\ layer$
Optimizer Adam
Dropout rate as 0.3;
End for
Normalize x
$initilize\ sequence\ to\ fixed\ length$
$f_p = \sigma(W_f x_t + U_f + b_f).$
$o_t = \sigma(W_o x_t + U_o + b_o)$
$Concat = [\overrightarrow{h_t}; \overleftarrow{h_t}]$
For each head **do**
$Q = HW^Q, K = HW^K, V = HW^V$
$MHA(Q, K, V) = Concat(head_1, \ldots head_h).W_o$
Add residual connection RC;
Layer normalization;
Calculate temporal aggression
$RC_{avg} = \frac{1}{T}\widehat{RC}$
Final activation SoftMax over 7 classes function for classification
End for
Collect loss;
$Loss = -\sum_{c=1}^{7} w_c y_c log\ (\hat{y}_c)$
Use optimizer to update weights
Evaluate final classification
$Predicted\ class = \arg\max(\hat{y})$
Attack detected map to one of seven traffic categories
Else
Normal
End if
End Algorithm

5 Experimentation, results and analysis

The experiments use a simple Bi-LSTM and transformer-based MHA with a small input layer to handle the first sequence, then one Bi-LSTM layer for tasks that do not need much computing power, which is good for devices with limited resources (i.e., 6G edge nodes), and a transformer-based MHA with fewer heads to understand long sequence relationships [48].

Unlike existing standalone transformer and deep learning models, which are hard to train because of their structural complexity, the suggested lightweight ISA model offers a simpler design, reduced computational complexity and efficient prediction performance, making it suitable for deployment on resource-constrained edge devices. This makes it easier to use on devices with limited memory and helps it predict results more efficiently [49]; the following section explains the experimental setup used for testing and analyzing the results based on model performance.

https://doi.org/10.1186/s13638-026-02676-9

Table 2 Experimental setup

Environment	Google colab
Libraries	Python 3.11, Tensor lite, scikit, NumPy
Model	Lightweight Bi-LSTM with MHA
Data split	70:30
Optimizer	Adam
Loss	Class weighted sparse categorical cross-entropy

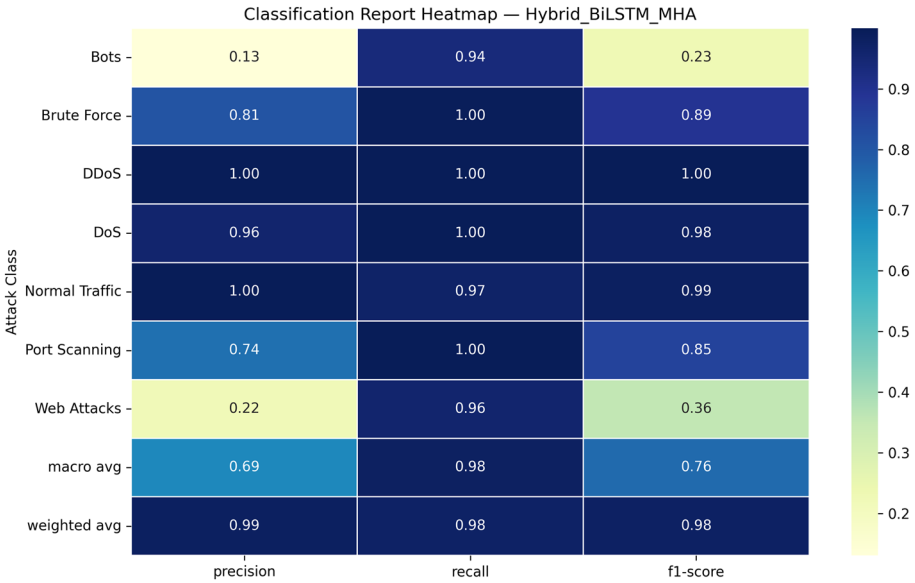


Fig. 3 Heatmap Representation of multiclass classification performance. The figure illustrates the heatmap of the classification metrics obtained from each attack category using proposed Bi-LSTM with Multi-head attention model. Performance indicators include precision, recall, F1 score and overall support are represented through color intensity, enabling rapid comparison across different attack classes. Darker region indicates stronger predictive performance, whereas lighter region identifies comparatively challenging attack categories

5.1 Experimental setup

The experimental setup of the proposed model is designed to detect various web attacks, including brute force, SQL injection, and shell scripting, using the dataset. The types of hardware and software used for training and testing are defined here. The system, which includes an Intel Core i7 processor, 16 GB of RAM, and an optional GPU, was responsible for processing the data during testing. Table 2 represents the experimental setup.

For testing and training, the work is done with Google Colab with a T4 GPU to handle sufficient training to predict web attacks that are processed inside the mobile communication network. This allows the heavy lifting of training the intellectual security architecture (ISA) model to be performed efficiently before its lightweight deployment in resource-constrained 6G edge environments.

5.2 Results

The proposed method, the intellectual security architecture (ISA), uses the Bi-LSTM with multi-head attention model to predict which type of web attack happens inside the mobile communication network. Figure 3 illustrates the heatmap for different types of web attack prediction.

The confusion matrix of this research represents which type of attacks are correctly identified by the model, like how many samples are tested and trained for getting proper classification. The proposed model, the intellectual security architecture (ISA), is tested with 500,000 samples (fig. 4), where it predicts that there are different web attacks like brute force attacks, SQL injection attacks, and shell script attacks in the mobile communication network, using a dataset split of 70% for training and 30% for testing class weightage with focal loss.

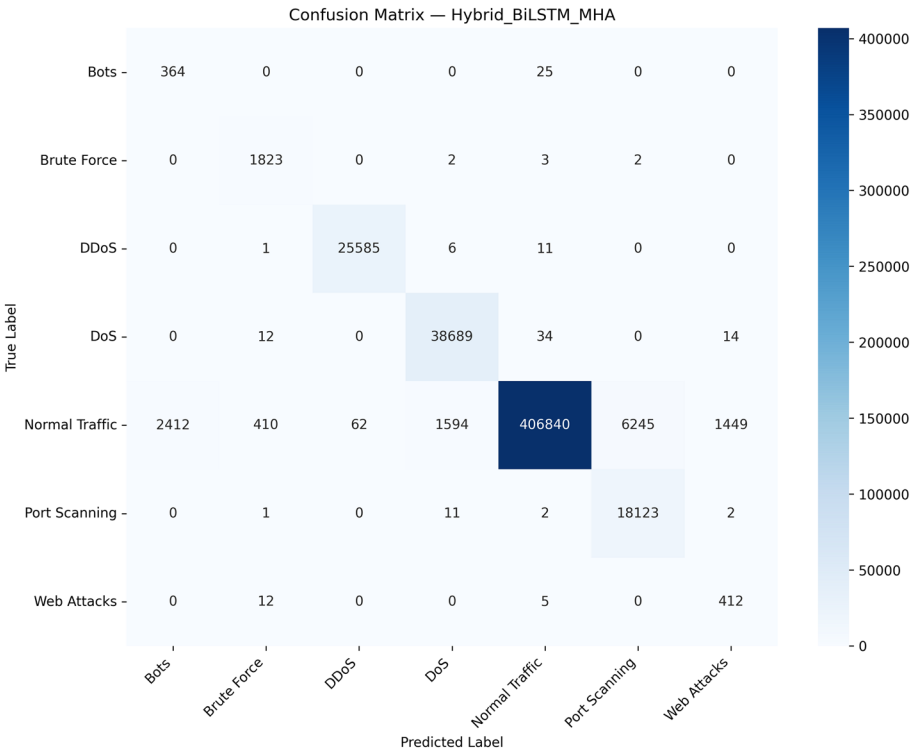


Fig. 4 Confusion Matrix of Multi-class attack prediction. The figure illustrates the confusion matrix generated by the proposed Bi-LSTM with multi-head attention classifier during multiclass web attack detection. Each row represents the actual attack category while each column corresponds to the predicted class. The concentration of values along the principal diagonal indicates correct classification whereas off-diagonal entries represent misclassified samples. The limited number of misclassifications demonstrates that the proposed attention enhanced learning framework effectively distinguishes between similar attacks patterns while maintaining high prediction accuracy across diverse web attack categories

The multiclass predicted classification is split into two classes: One is predicted and another is true value. The proposed model, the intellectual security architecture (ISA), predicts various types of attacks with high efficacy in the 6G edge communication network, achieving a detection accuracy of 97% and an AUC score of 98%.

5.3 Performance metrics

The proposed Bi-LSTM and multi-head attention lightweight model, which forms the core of the intellectual security architecture (ISA), was evaluated using standard classification metrics, including accuracy, AUC score, and confusion matrix analysis. The performance was assessed across multiple attack classes represented in the CICIDS2017 dataset, with particular focus on web attack detection capabilities within edge-based content sharing environments.

5.3.1 Model accuracy and loss

Significant insights of learning and efficacy are unfolded in terms of web attack detection through the optimization trajectory of the hybrid Bi-LSTM and multi-head attention model (ISA). Baseline accuracy of approximately 85%, including both training and validation of datasets, is what is exhibited by the model with regard to performance improvement over 50 training epochs, demonstrating effective learning for edge environments. Figure 5 plots the accuracy and loss model and mathematical equation is expressed in Eq. (22).

$$\text{Accuracy} = \frac{\text{TN} + \text{TP}}{\text{TN} + \text{FN} + \text{TP} + \text{FP}} \tag{22}$$

The proposed model (ISA) maintains steady improvements in accuracy across all epochs, and by epoch 50, it reaches approximately 97% accuracy. While the validation accuracy shows greater variability, overall it tends to

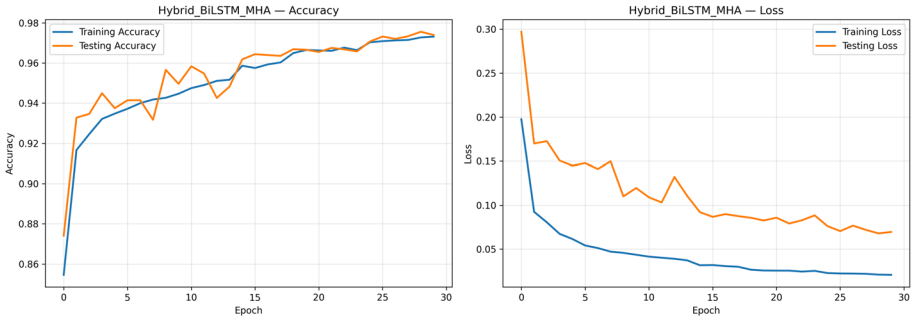


Fig. 5 Training and validation accuracy and loss curves of the proposed model. The figure presents the learning behavior of the proposed model throughout the training process. The accuracy curve demonstrates a consistent improvement in prediction performance for both training and validation dataset across successive epochs while corresponding loss curve gradually decrease before converging. The results confirm that the proposed optimization strategy enables effective feature learning and reliable generalization for multiclass web attack detection

follow a similar upward trajectory. The model crosses a critical threshold in recognizing patterns of web attack detection during epochs 45–50, as the validation accuracy jumps from 80 to 97%.

The loss value provides a complementary view of the learning process of the model. Although the initial loss begins at a very high value of around 1.4, the training loss remains near zero consistently. The significant difference between training and validation loss suggests some overfitting, despite the high validation accuracy. The model exhibits generalization capability, as during training the validation loss decreases substantially to 0.3 until the final epoch.

The model performs best around epoch 8, where the validation accuracy reaches about 97% before dipping slightly in the last epoch. This pattern and the stabilization of the validation loss at epoch 7 advocate diminishing returns of training after this epoch. The training results back up the performance measurements, showing that the hybrid Bi-LSTM and multi-head Attention architecture (ISA) is effective in dealing with different web attacks in mobile networks and 6G IIoT edge environments.

5.3.2 AUC

The process is to separate the class of each attack that is performed inside the mobile communication network and this is most useful in imbalanced dataset. Figure 6 displays the AUC curve of the proposed work.

5.3.3 Stochastic neighbor embedding

The T-SNE (t-distributed stochastic neighbor embedding) is a method to reduce dimensionality in the visualization of multiclass attack prediction. The initial data from the prediction are processed from 2D as high-dimensional data into low-dimensional data to visualize all attack classes for preserving the local structure.

The multiclass visualization, presented in cluster form, distinguishes between normal and attack classes while capturing nonlinear relationships. This is vital for validating the discriminative power of the Intellectual Security Architecture (ISA) in separating different attack types within 6G network traffic. Figure 7 illustrates how the proposed model effectively separates different attack classes through proper representation.

5.4 Ablation study

To verify the accuracy gains from the amalgamation of Bi-LSTM and MHA rather than from either standalone component, both branches were trained and evaluated independently under identical preprocessing configurations, class weightings, and training schedules.

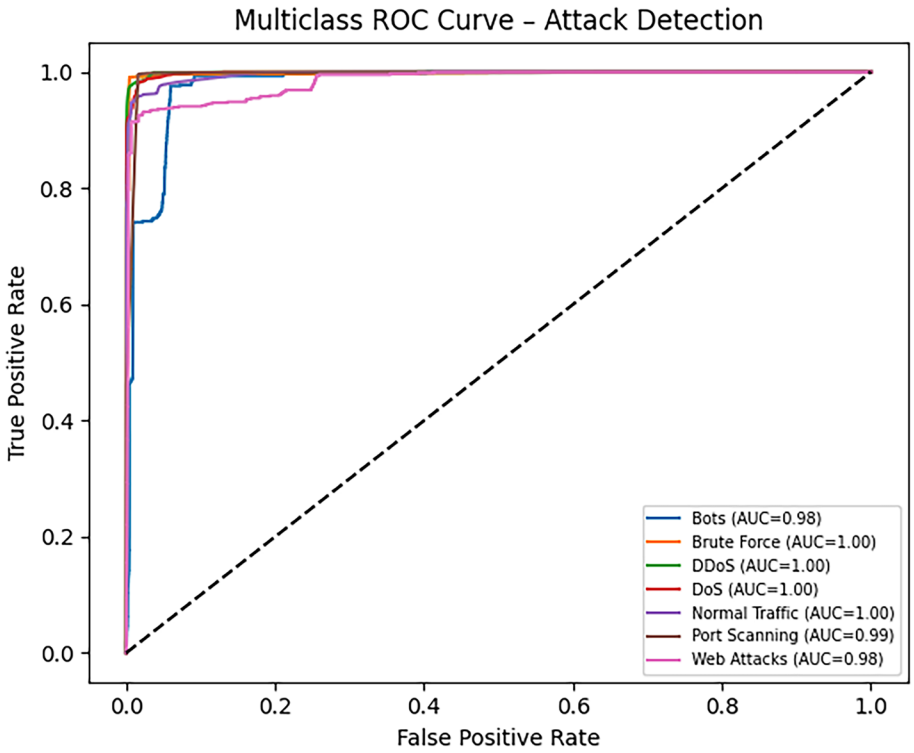


Fig. 6 Receiver operating characteristic (ROC) and area under curve (AUC) analysis for multiclass attack detection. The ROC curve illustrates the discrimination capability of the proposed Bi-LSTM with multi-head attention framework across multiple web attack categories. Each curve represents the trade-off between the true positive rate and false positive rate of the specific attack class under varying decision thresholds. The high AUC values confirm that the proposed attention guided learning framework effectively separates legitimate traffic from multiple categories of malicious network activities

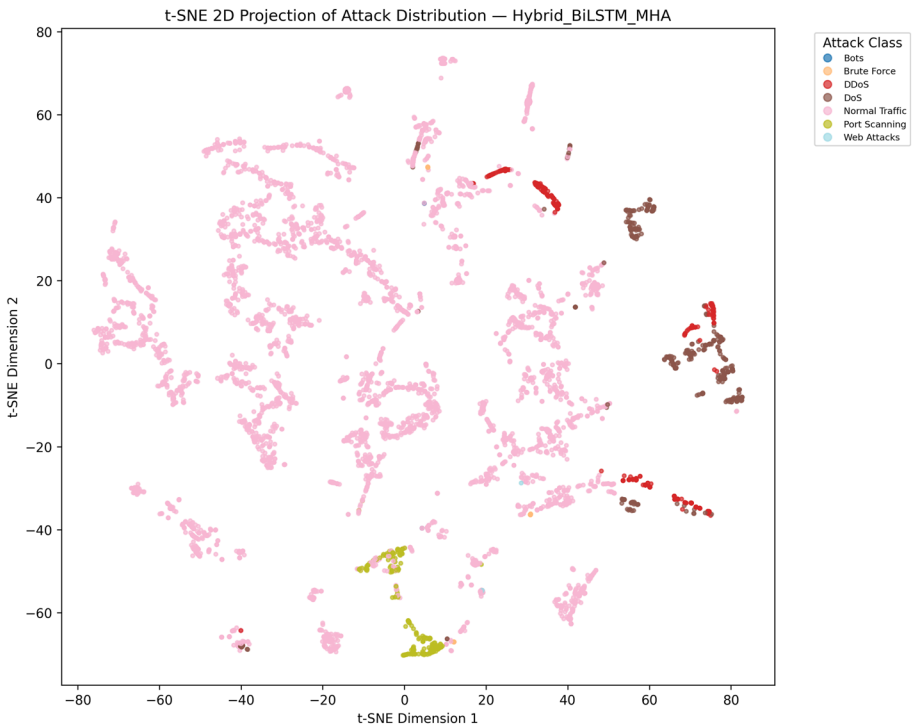


Fig. 7 Two-Dimensional t-SNE visualization of learned attack feature distributions. The figure illustrates the 2D t-distributed stochastic neighbor embeddings (t-SNE) projection of feature representations extracted by the proposed Bi-LSTM with multi-head attention network. Each point represents an individual traffic samples projected into a lower dimensional feature space while preserving neighborhood relationships. Distinct clusters correspond to different web attack categories; the clear cluster boundaries indicated reduced feature overlap and improved class discrimination, contributing to higher multiclass classification performance

From Table 3, the MHA branch performs lower than other components as 91,845 accuracy confirms that temporal sequence modeling is the dominant attribute on this dataset. The Bi-LSTM component only archives 97.50% accuracy and hybrid ISA architecture has consistent improvement as 97.56%, matching F1 of 0.9791 vs 0.9786 these improvements are incremental on CICIDS2017 dataset. The larger practical value is shown under

Table 3 Ablation study differentiating hybrid contribution

Model	Accuracy	Precision	Recall	F1	AUC
Bi-LSTM only	0.9750	0.9850	0.9750	0.9786	0.9991
MHA only	0.9184	0.9541	0.9184	0.9316	0.9865
Hybrid Bi-LSTM+MHA	0.9756	0.9854	0.9756	0.9791	0.9991

distribution shift on Sect. 4.6 where the hybrid model outperforms Bi-LSTM component only with 60.59% vs 59.22% accuracy, 0.9324 vs 0.9279 AUC on Edge IIoT 6G node network behavior dataset.

5.5 System level evaluation

The system level evaluation is necessary for 6G edge deployment feasibility because 6G edge nodes operate under rigid compute, memory, and energy constraints, while deploy ability cannot be evaluated solely on classification accuracy. Table 4 defines parameter count, inference latency, and throughput of all three variants on same hardware.

The full hybrid model requires under half a MB of storage and adds only ~2.4 ms median latency over the Bi-LSTM component with cost for the accuracy and cross domain robustness shown in Sect. 4.6. This lightweight footprint is what makes it realistic to deploy directly on edge hardware and fine-tuned rather than requiring centralized retraining per deployment site. The measured process memory as ~3.4 GB reflects full evaluation footprint data loader not the ISA architecture memory. A production-based edge deployment would use a quantized runtime with TF-lite as lower footprint. This runtime will explicitly avoid an apparent difference between 30 k parameters and 3.4gb process memory.

5.6 Cross dataset evaluation

Table 5 reports the strict zero shot performance on edge IIoT dataset without fine tuning. The classification accuracy decreases relative to CICIDS2017 (97.56% to 60.59%) under strict zero shot transfer dataset with different feature distribution and web attack taxonomy with the near total absence of cross-dataset testing in existing hybrid IDS literature. Critically, AUC remains high at 0.93%, and the throughput as 11.52 s which indicates that the model continues to rank web attack traffic with strong separability, even without domain adaptation. An edge deployment can use the model output score with locally calibrated threshold a lightweight on device step to recover much of the detection utility the AUC indicates while footprint defined in Sect. 4.5 makes locally federated fine tuning computationally realistic at the edge deployment.

Table 4 System level evaluation

Model	Parameters	Disk size (MB)	Median latency (ms)	P95 latency (ms)	Throughput (samples/s)
Bi-LSTM only	13,319	0.197	85.86	94.77	11.65
MHA only	21,511	0.306	87.13	96.21	11.48
Hybrid Bi-LSTM+MHA	30,087	0.418	88.24	96.95	11.33

Table 5 Cross dataset evaluation

Model	Accuracy	Precision	Recall	F1	AUC	Throughput samples/S
Bi-LSTM only	0.5922	0.6581	0.5922	0.5944	0.9279	11.52
MHA only	0.4412	0.4962	0.4412	0.4329	0.8721	11.49
Hybrid Bi-LSTM+MHA	0.6059	0.6798	0.6059	0.6070	0.9324	11.52

Table 6 Proposed model web attack detection compared with another model

Model	Multi class web attack classification accuracy %	Attack detected
Proposed Bi-LSTM with MHA	97	Brute force, shell script and SQL injection
Transformer MHA	91.84	BOTS
Cat-Boost	88	SQL injection
Char-LSTM	85	Service attacks
CNN-LSTM	95.5	XSS attack
Support vector machine	91	SQL injection
Naïve Bais	94	SQL injection

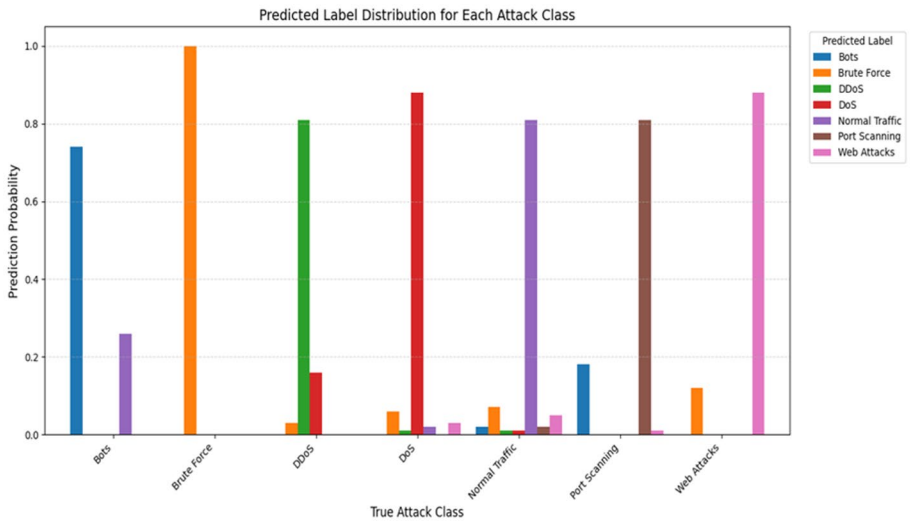


Fig. 8 Visualization of predicted web attack categories generated by the proposed framework. The figure illustrates the prediction outcomes produced by the proposed Bi-LSTM with multi-head attention classifier for different categories of web attacks. Each prediction corresponds to the model final classification after sequential feature extraction and attention-based feature refinement. The visualizations demonstrate the model's capability to correctly identify multiple attacks classes while minimizing false predictions

Table 6 compares the proposed model against classical machine learning and standalone deep learning baselined evaluated under identical data preprocessing and finally a comparison against transformer-based IDS is added as benchmarking to define how well the proposed model work effectively in predicting web attacks on Edge environments.

6 Discussion

The results of the suggested method, Bi-LSTM with MHA, which forms the core of the intellectual security architecture (ISA), accurately predicts different types of web attacks, achieving a 97% detection rate and an AUC score of 98%, which demonstrates its effectiveness compared to some current deep learning models.

The main advantage of this research is to propose a lightweight model that supports edge device integration in 6G networks without the need for additional resources. The efficiency of the model is high compared to other existing models, as well as in memory-constrained devices. Figure 8 shows the final predicted value of attacks from the dataset sample.

The model is excellent at accurately recognizing harmless network traffic with very few mistakes, as shown by the darkest cell in the top-left corner of the heatmap. This functionality is particularly important in real-world 6G edge deployment scenarios to minimize false alarms and maintain quality of service.

Light shades of off-diagonal elements show the misclassification of different attack types. The heatmap indicates that there is confusion between the brute force web and shell scripting attack categories, although this confusion occurs at relatively low rates of 3–5%. Due to diverse implementation patterns, classification accuracy is scarce, particularly for SQL injection attacks. In this respect, the proposed hybrid architecture (ISA) shows a significant classification improvement compared to traditional ML and conventional DL models in securing mobile communication networks.

Figure 9 illustrates the accuracy comparison of Bi-LSTM with MHA (ISA) over baseline models. Table 6 defines the web attacks detected by the proposed and traditional model. This comparison highlights the superior performance of the intellectual security architecture for securing 6G edge-based content sharing.

Table 6 compares the proposed model against classical machine learning and standalone deep learning baselined evaluated under identical data preprocessing, and finally, a comparison against transformer-based IDS is added as benchmarking to define how well the proposed model work effectively in predicting web attacks on edge environments.

Sections 4.5 and 4.6 provides support for a bounded 6G applicability claim rather than presenting a generalized claim. The model is lightweight and fast enough to run on constrained edge hardware with 30,087 parameters, less than 0.5 MB memory footprint and median latency less than 90 ms. The proposed model is trained and preserved strong ranking-based attack versus normal separability (AUC as 0.93%) even applied with zero shot to an unseen edge IIoT traffic distribution. The accuracy reduction observed under domain shift indicates direct deployment across diverse 6G traffic environments. The low computational cost of the model helps in realistic threshold calibration and fine tuning that in return yields low-cost path that closes the performance gap. This proposed ISA architecture offers a more defensible claim of edge device suitability and real-time deployment potential asserting deployment readiness in its current zero shot form.

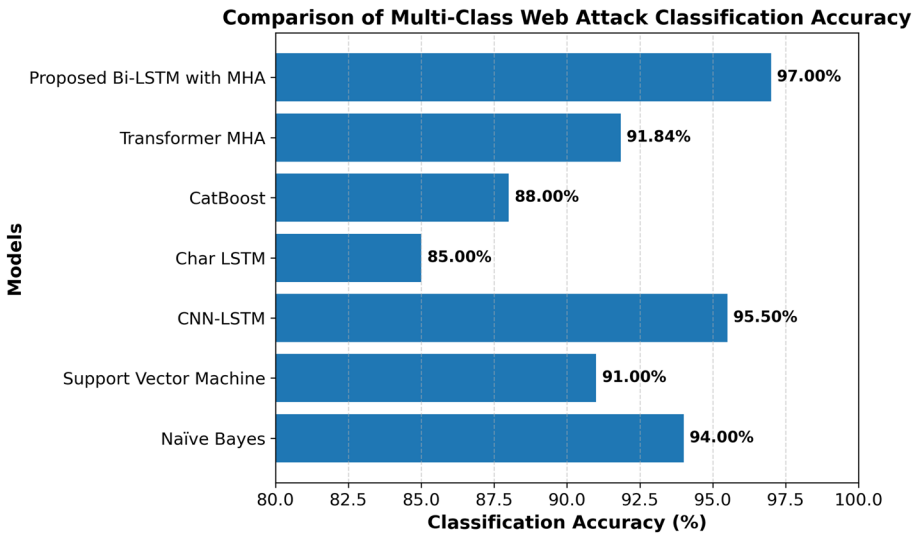


Fig. 9 Comparative performance evaluation of the proposed Bi-LSTM with multi-head attention model against existing methods. The figure compares the proposed Bi-LSTM with multi-head attention framework against several conventional machine learning and deep learning approaches using standard performance metrics such as accuracy. The proposed model consistently achieves superior classification performance due to its ability to simultaneously capture long-term sequential dependencies and emphasize the most informative traffic feature through attention mechanism. The comparative analysis shows that the integration of multi-head attention with Bi-LSTM enhances multiclass web attack detection performance while maintaining improved robustness and generalization compared with existing baseline models

7 Conclusion and future work

The proposed ISA demonstrates strong in distribution detection performance on the CICIDS2017 dataset. With a resource footprint of 30,087 parameters, and sub 90-ms inference time, the model is highly compatible with 6G edge hardware constraints. Zero shot evaluation on edge IIoT shows web attack ranking capability as AUC 0.93% generalizes beyond its training distribution even without adaption while classification accuracy under this shift highlights the need for lightweight on device federated fine-tuned before the model is considered deployment ready across heterogeneous 6G edge environments. The hybrid architecture offers a significant improvement over the traditional models. The system demonstrates high classification accuracy, particularly for web attacks; it achieves approximately 97% and effectively identifies benign traffic with minimal false positives. The better performance of the model shows that the Bi-LSTM and multi-head attention design work well for telling apart different web attack patterns in mobile networks. This shows that the suggested model is better at predicting various web attacks on uneven datasets with few resources and can be easily used on edge devices to monitor and send security alerts about attacks.

In the future, the intent is to include a federated learning model using FedAvg Protocol, where edge device trains local copy of the model on its own traffic and weight updates are transferred to central aggregator every k rounds directly targeting the generalization. After training INT8 quantization and pruning structured will be applied using harness quantifying the accuracy trade of explicitly target less than 50% current parameter count with less than 2-point accuracy loss. SHAP values will be computed over the output layer to attribute predictions to specific traffic features like packet length, flow duration, and destination port to test MHA attention weights align with SHAP-identified important features. Finally, a zero-day detection using unsupervised auto encoder branch trained on normal traffic using reconstruction error as an anomaly score will provide a complementary zero-day detection path alongside the supervised classifier.

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Data availability Authors declare that all the data being used in the design and production cum layout of the manuscript is declared in the manuscript and dataset used in the research https://www.kaggle.com/datasets/ericanacletoribeiro/cicids2017-cleaned-and-preprocessed and https://www.kaggle.com/datasets/mohamedamineferrag/edgeiiotset-cyber-security-dataset-of-iot-iiot. Code Available Online: https://github.com/TECHcode317/ISA-ABLATION-and-Cross-validation.

Declarations

Competing interests The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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